Quality Analysis of Basmati Rice



A.D.D.S.De Silva (S/18/338)

Digital Image Processing (CS 314)

Content

01 Introduction to problem and the proposed system

02 Approach

03 Procedure

04 Quality Analysis based on measurements

05 Conclusion



Introduction to the problem and proposed System

01 **Rice one of main food source**

02 **Manual Inspection :**
Time consuming, costly, inefficient

03 Proposed System: Image processing Techniques

04 Based On: Length of rice grain, Aspect ratio, Chalky, Broken Seeds, impurity Content

Process

Image Acquisition

Image Pre processing

Image Segmentation

Feature Selection and Extraction

Classification and Decision

Data Set













Approach

- Step 01: Image preprocessing and noise removal
- Step 02: Apply Shrinkage algorithms for segmentation
- Step 03: Perform edge detection
- Step 04: Seed Measurements
- Step 05: Classify based on Measurements

Approach and Procedure

1. Image Preprocessing and Image Segmentation

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

FONT_SIZE = 8
COLOR = 'maroon'

# Loading the images using cv2.imread() function
img_bgr = cv2.imread("Images/image_01.jpg", 1)
img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)
img = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2GRAY)
# Applying median filtering
median_filtered = cv2.medianBlur(img, 5)
# Applying threshold algorithm to segement rice grains from the black background
ret, thresh = cv2.threshold(median_filtered, 160, 255, cv2.THRESH_BINARY)
```

3. Applying Morphological Operations

```
# Applying erosion to seperate the touching features of rice grains
erosion = cv2.erode(dst, kernel2, iterations=3)
# Apply dilation to grow the eroded features back to their original shape
dilation = cv2.dilate(erosion, kernel2, iterations=1)

# Apply edge detection algorithm  to find the region of boundaries of rice grains
edges = cv2.Canny(dilation, 100, 200)
```

3.Feature Extraction and Feature Selection: count, major axis length, minor axis length

```
# object Measurement
contours, hierarchy = cv2.findContours(erosion, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
num_seeds = len(contours)

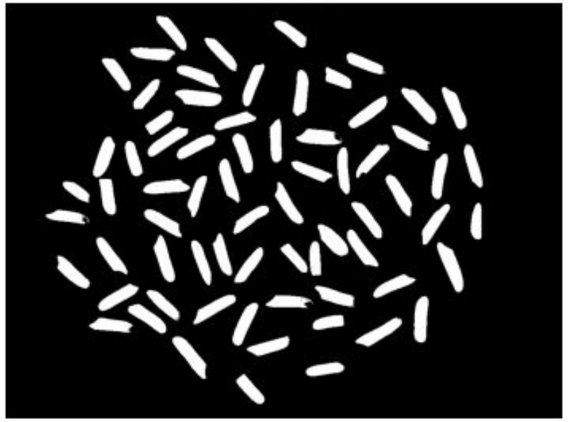
#Calculating major axis, minor axis and calculating ratio
for idx,contour in enumerate(contours):
    seed_number = num_seeds - idx
    x, y, w, h = cv2.boundingRect(contour)
    aspect_ratio = float(w)/h

    major_axis = w
    minor_axis = h
```

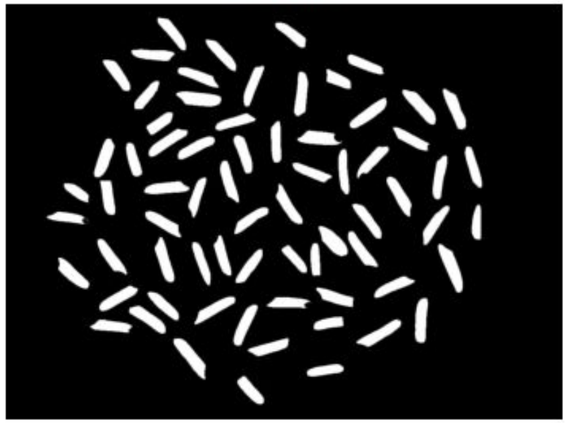
Original Image



Binary Image



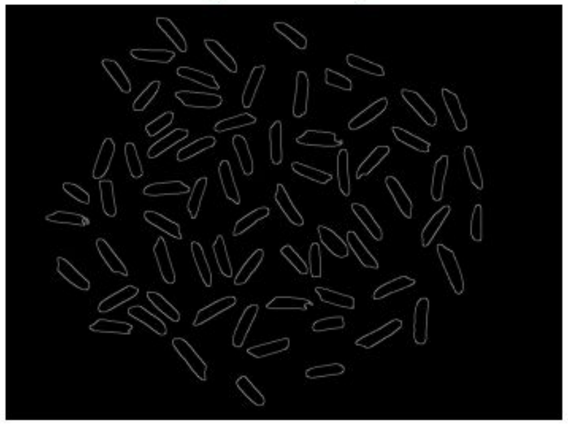
Eroded Image



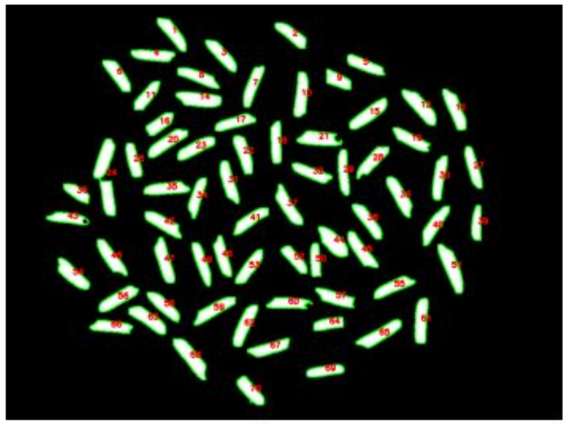
Dilated Image



edge detected Image



Contourded

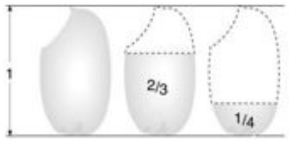


3.Classification and Decision

- **Based on Major Axis Length**
- **Based on Aspect Ratio**
- **Based on Chalky Content**
- **Based on Broken seed content**
- **Based on Impurity Content**

Based on Major Axis Length	**Quality Measurements**	**Length (pixels)**
	Very Long Grain	Above 42
	Long Grain	42 - 44
	Medium Grain	30 - 42
	Small Grain	26 - 30
Based on Aspect Ratio	**Quality Measurements**	**L/W ratio**
	Slender	Over 3.0
	Medium	2.1 – 3.0
	Bold	1.1 – 2.0
	Round	1.0 or less
Based on Chalky Degree	**Quality Measurements**	**Chalky Degree (%)**
	Affected Rice	>0.3 (30%)
	Good Rice	<0.3 (30%)
Based on Broken Rice	**Quality Measurements**	**Broken Percentage (%)**
	Very Good Quality	<= 4.90
	Good Quality	<= 19.75
	Low Quality	<= 34.50
	Very Bad Quality	<= 40.00

Standards: International Rice Research Institute.



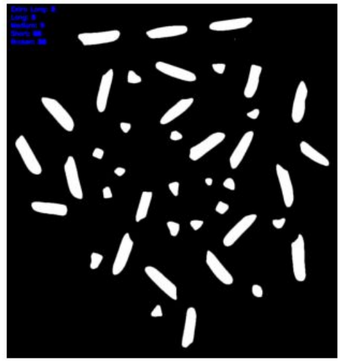
Broken Rice

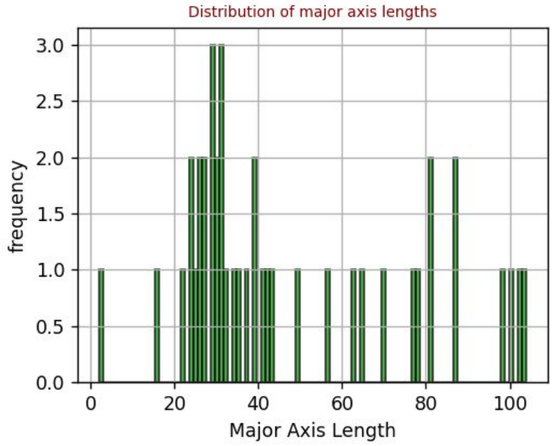


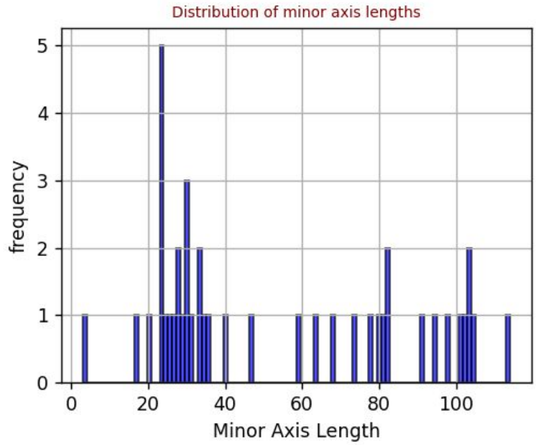
Chalky Rice

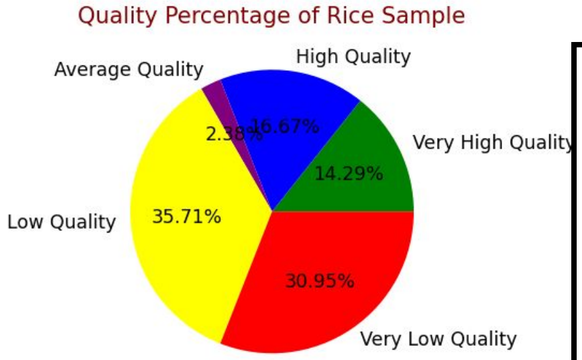
Based on Major Axis Length









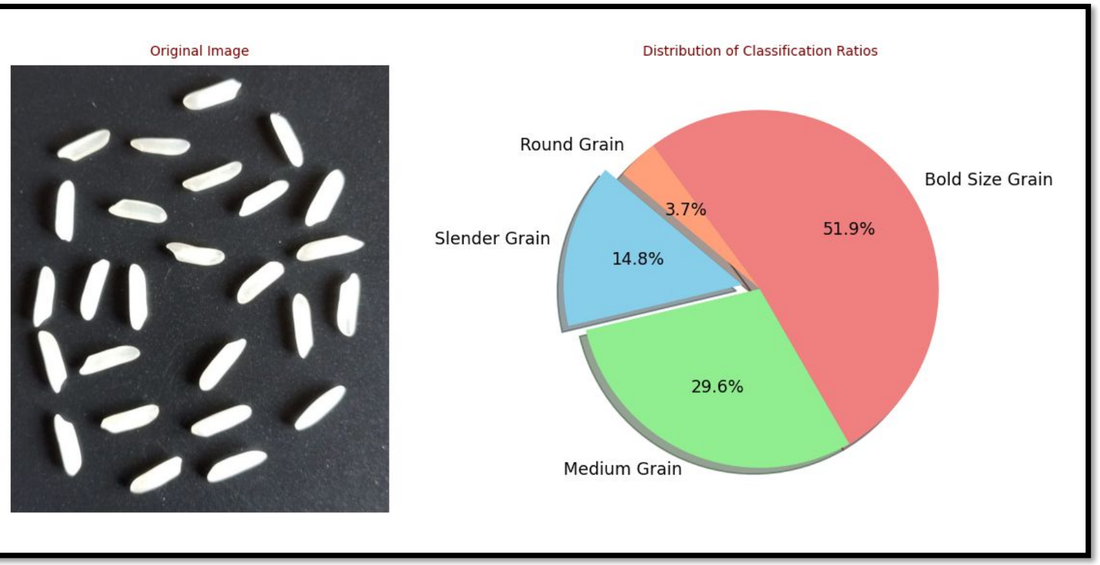



```
********************Counts ********************

No of rice grains =  42
Count of Extra Long Grain: 6
Count of Long Grain: 7
Count of Medium Grain: 1
Count of Short Grain: 15
Count of Broken Grain: 13

**********Analysis based on the major axis length of the rice granules***
14.29% of Very High Quality Rice
16.67% of High Quality Rice
2.38% of  Average Quality Rice
35.71% of Low Quality Rice
30.95% of Very Low Quality Rice
```

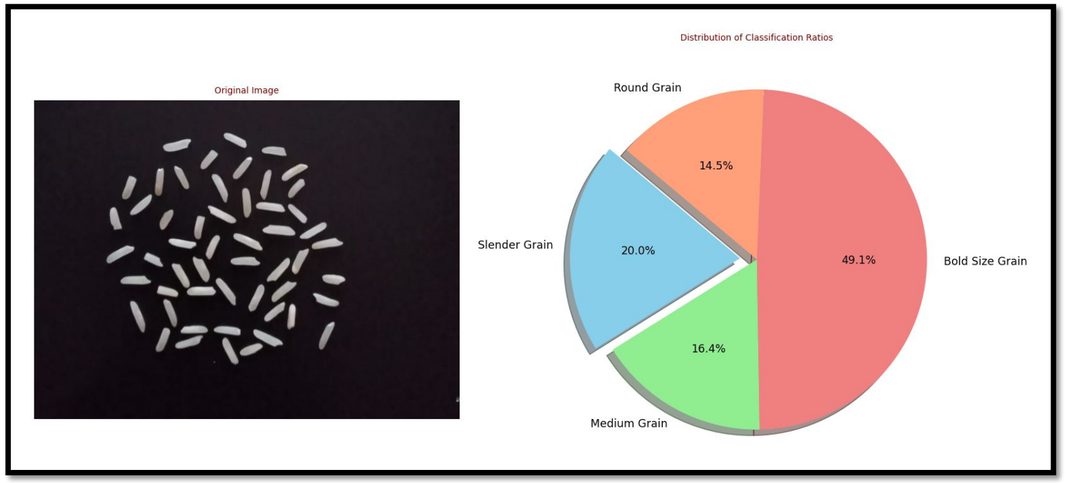
Based on Aspect Ratio




```
******************Count******************
No of rice grains =  27
Count of Slender Grain: 4
Count of Medium Grain: 8
Count of Bold Size Grain: 14
Count of Round Grain: 1

**********Percentage of each category *********
Percentage of Slender Grain: 14.814814814814813
Percentage of Medium Grain: 29.629629629629626
Percentage of Bold Size Grain: 51.85185185185185
Percentage of Round Grain: 3.7037037037037033

**********Analysis based on the aspect ratio of the rice granules****
14.81% of Very High Quality Rice
29.63% of High Quality Rice
51.85% of Average Quality Rice
3.70% of Low Quality Rice
```




```
******************Count******************
No of rice grains =  55
Count of Slender Grain: 11
Count of Medium Grain: 9
Count of Bold Size Grain: 27
Count of Round Grain: 8

**********Percentage of each category *********
Percentage of Slender Grain: 20.0
Percentage of Medium Grain: 16.363636363636363
Percentage of Bold Size Grain: 49.09090909090909
Percentage of Round Grain: 14.545454545454545

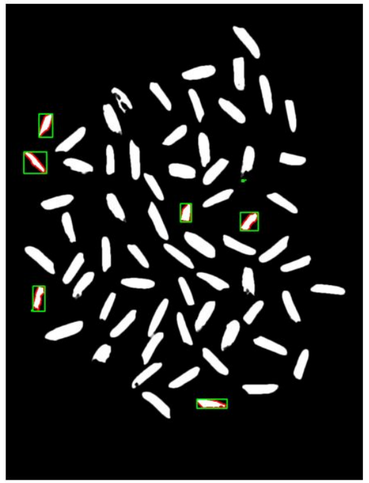
**********Analysis based on the aspect ratio of the rice granules***
20.00% of Very High Quality Rice
16.36% of High Quality Rice
49.09% of Average Quality Rice
14.55% of Low Quality Rice
```

Based on Broken Seed Content

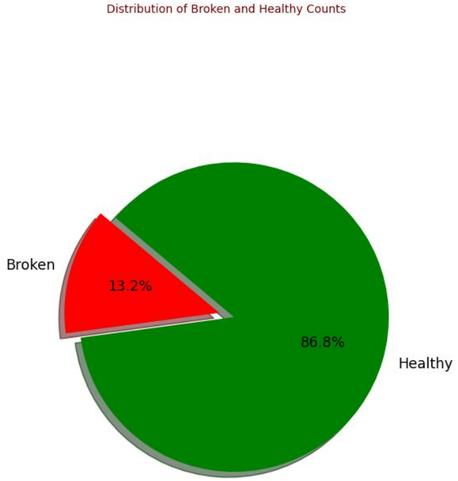
Rice Sample



Broken Seed Detection



Distribution of Broken and Healthy Counts

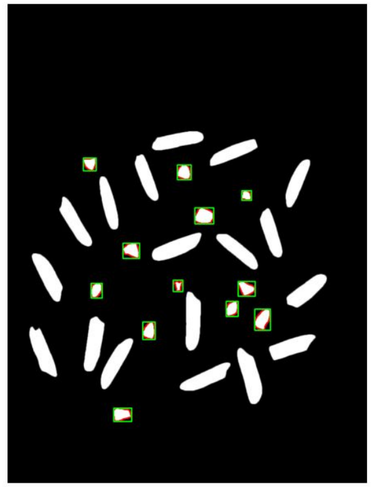


Quality of Rice : Good Quality

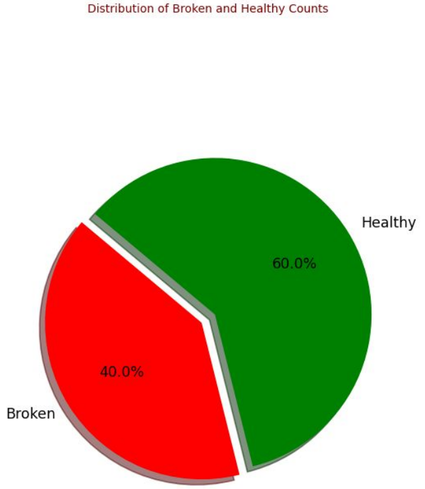
Rice Sample



Broken Seed Detection



Distribution of Broken and Healthy Counts



Quality of Rice : Very Bad Quality

Based on Chalky Content

Rice Sample



Chalky Detection



Chalkiness Percentage: 85.39%
Chalkiness Level: Affected

Used methods

- **Extended Maxima Operator**

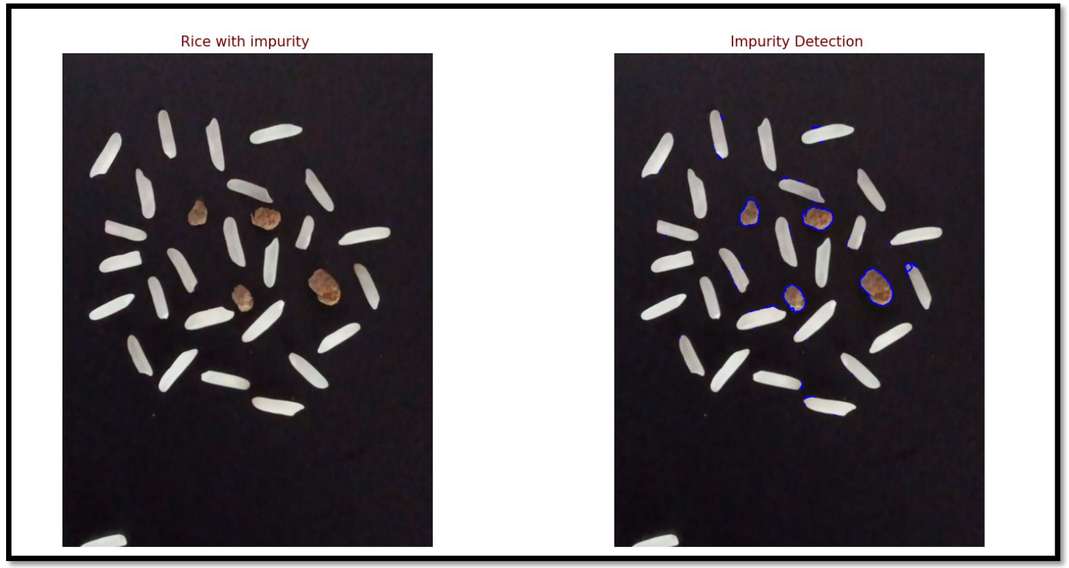
```
# Create a structuring element for the extended maxima transform
kernel = np.ones((3,3), np.uint8)
# Apply Canny edge detector to preprocess the image
edges = cv2.Canny(image, 100, 200)
# Apply morphological reconstruction
reconstruction = cv2.dilate(edges, kernel, iterations = 1)
# Apply extended maxima transform
extended_maxima = reconstruction - edges
# Calculate the total area of chalky regions
chalky_area = np.sum(extended_maxima == 255)
```

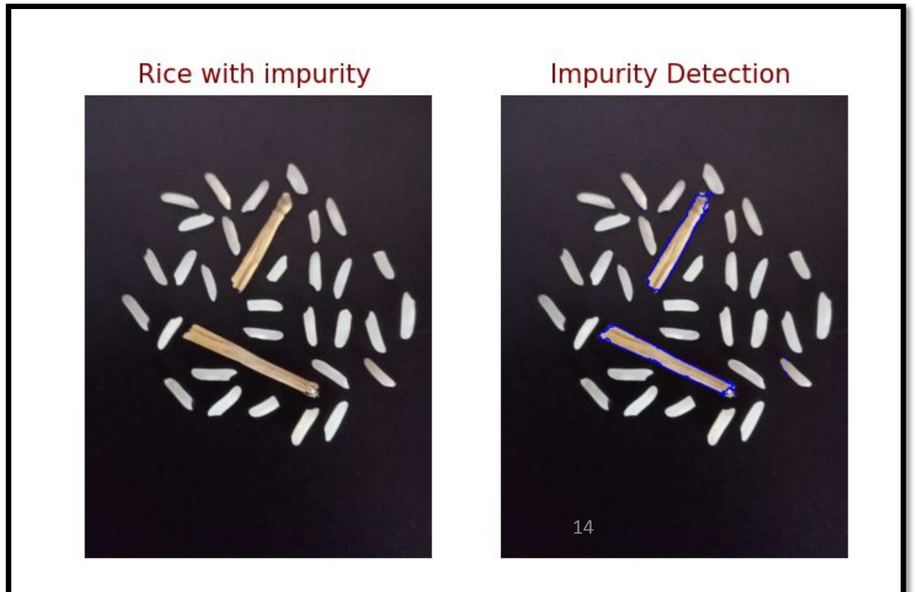
- **Thresholding**

```
# Apply thresholding
ret, thresh = cv2.threshold(image, 200, 255, cv2.THRESH_BINARY)
```

Based on Impurity Content







Original Image



```
No of rice grains =  70

**********Analysis based on the major axis length of the rice granules*****

0.00% of Very High Quality Rice
35.71% of High Quality Rice
35.71% of  Average Quality Rice
31.43% of Low Quality Rice
14.29% of Very Low Quality Rice

**********Analysis based on the aspect ratio of the rice granules**********

21.43% of Very High Quality Rice
35.71% of High Quality Rice
51.43% of Average Quality Rice
10.00% of Low Quality Rice

**********Analysis based on broken seed**********
Broken count:  7
Healthy count:  63
Broken percentage:  11.11 %
Quality of Rice :  Good Quality

**********Analysis based on the Chalky level**********
Percentage of chalky area: 5.78%
Quality: Good Quality

**********Analysis based on the impurity content**********
Percentage of impurity area: 0.71%
```

Original Image



```
No of rice grains =  26

**********Analysis based on the major axis length of the rice granules*******

26.92% of Very High Quality Rice
42.31% of High Quality Rice
26.92% of  Average Quality Rice
23.08% of Low Quality Rice
3.85% of Very Low Quality Rice

**********Analysis based on the aspect ratio of the rice granules**********

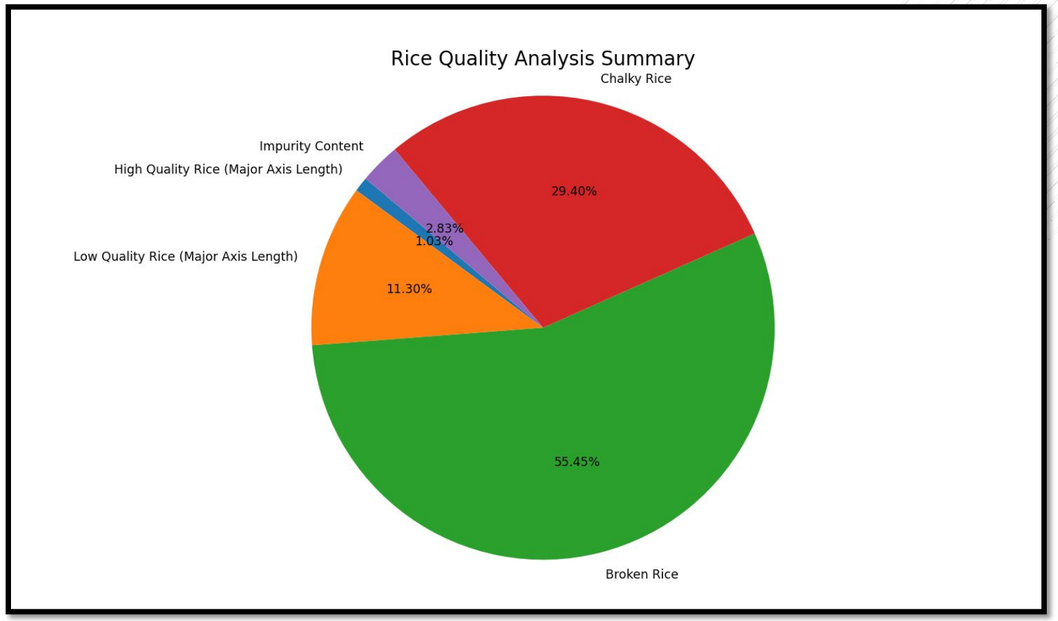
7.69% of Very High Quality Rice
26.92% of High Quality Rice
61.54% of Average Quality Rice
7.69% of Low Quality Rice

**********Analysis based on broken seed**********
Broken count:  1
Healthy count:  25
Broken percentage:  4.00 %
Quality of Rice :  Very Good Quality

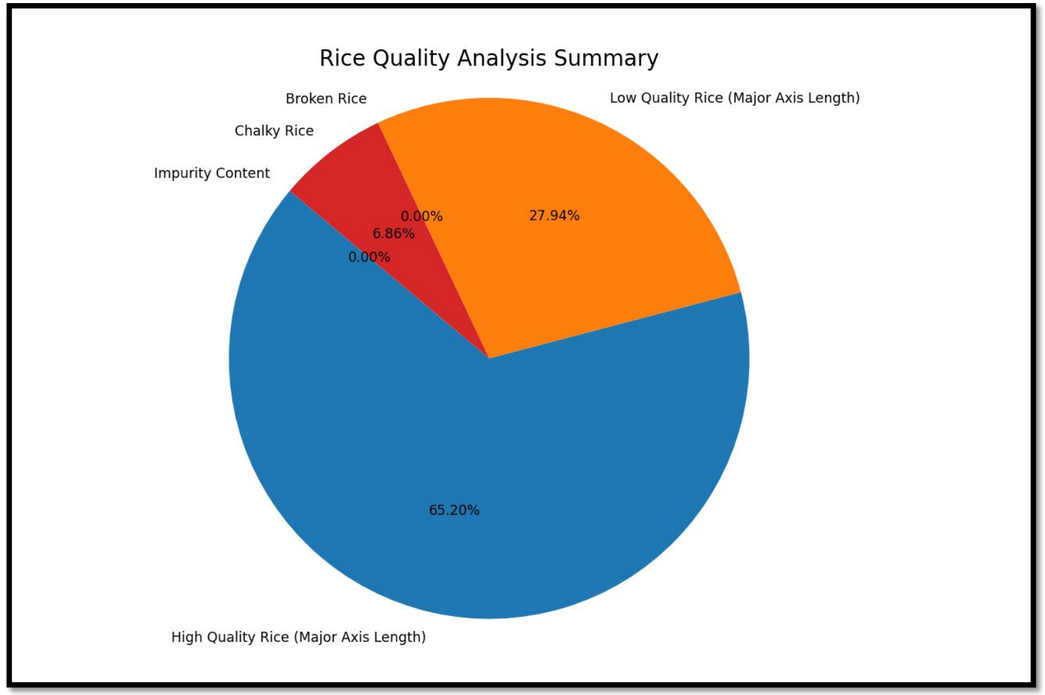
**********Analysis based on the Chalky level**********
Percentage of chalky area: 4.03%
Quality: Good Quality

**********Analysis based on the impurity content**********
Percentage of impurity area: 0.00%
```

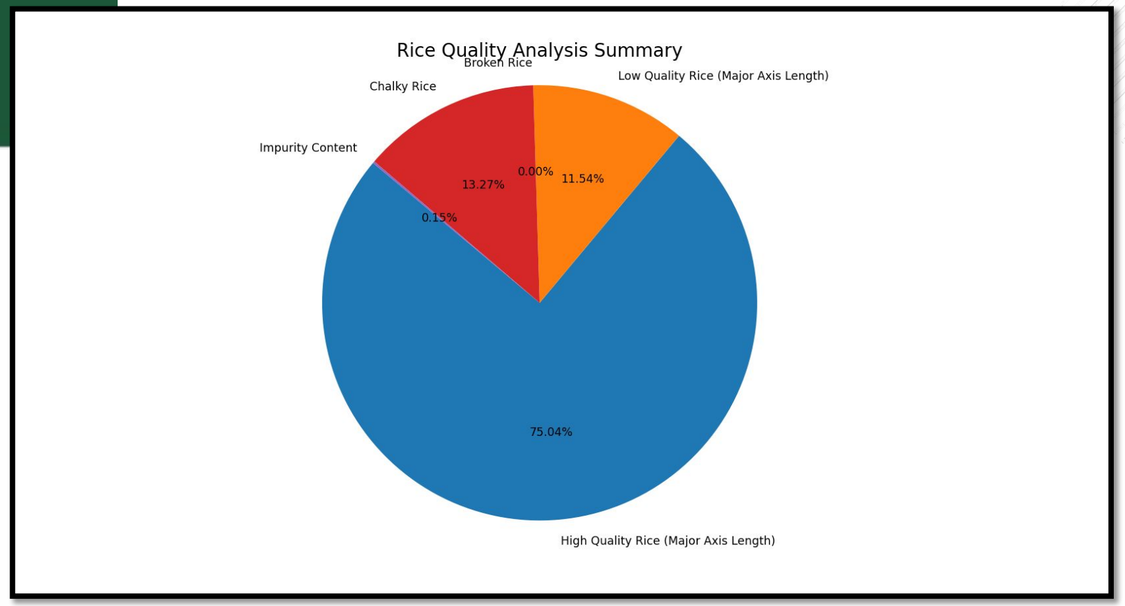



Rice Quality Analysis Summary
Chalky Rice
Impurity Content
High Quality Rice (Major Axis Length)
Low Quality Rice (Major Axis Length)
Broken Rice
29.40%
2.83%
1.03%
11.30%
55.45%

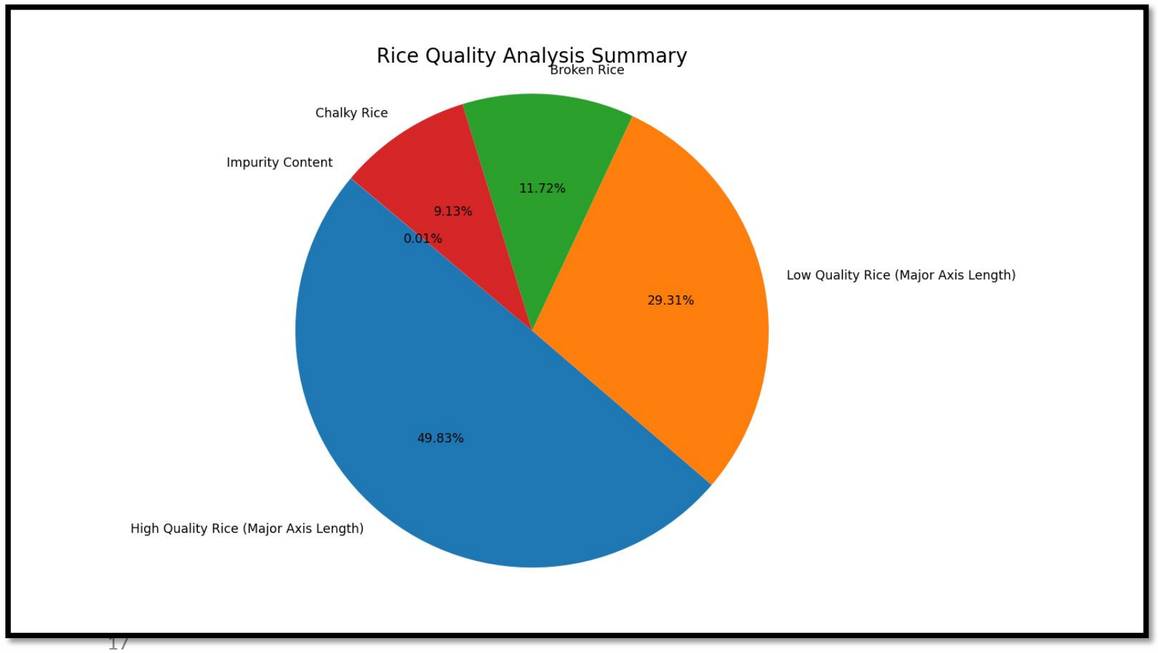



Rice Quality Analysis Summary
Broken Rice
Low Quality Rice (Major Axis Length)
Chalky Rice
Impurity Content
High Quality Rice (Major Axis Length)
0.00%
27.94%
6.86%
0.00%
65.20%




Rice Quality Analysis Summary
Broken Rice
Low Quality Rice (Major Axis Length)
Chalky Rice
Impurity Content
0.00%
11.54%
13.27%
0.15%
75.04%
High Quality Rice (Major Axis Length)


Original Image


Rice Quality Analysis Summary
Broken Rice
Chalky Rice
Impurity Content
11.72%
9.13%
0.01%
29.31%
Low Quality Rice (Major Axis Length)
49.83%
High Quality Rice (Major Axis Length)

After Analysis?

- **Very high quality rice**
  - Human Consumption
- **High Quality Rice (Second quality)**
  - Human Consumption
- **Broken Rice**
  - Not used for human Consumption
  - Mainly used for animal consumption in farms
- **Rice with high chalky content**
  - Highly fragile
  - Used for making rice flour
- **Rice with average quality of rice and with high chalky parts**
  - Used to make rice flour
- **Rice with very high amount of impurities**
  - Not used for consumption



Results and Discussion

- In this project classification is done based on many factors.
- Existing work only detect rice grains, count rice grains and considers only the length of the grain.

CONCLUSION

- Grading of rice can be effectively be done by using image processing techniques.

- Main benefits:
  - Require minimum time
  - Less cost
  - Gives better results than manual inspection method and traditional methods

THANK YOU

